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WEEK 2

(FED) 1.2

CSS 101
Box Model
Fonts

2025

ngee ann polytechnic

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LEARNING POINTS



- Name your files using lower-case (in web dev, files are hosted on the server. Having About.html about.html are treated differently by the server.)
- Name your files with intent and meaning (e.g Contact Us page should be contact-us.html and not page3.html
- Have spaces in your file names? Replace them with dashes "-" instead. This reduces conflict when people are typing and it's also easier to recognise.
- In web development, ALWAYS ensure that you have the proper file extension. For Windows users, it is highly recommended you enable file extensions in your system.
- Submission criteria -> Create a zip file of all your work. (services.html.. Home.html -> combine to form a zip file)
- Students mindset? -> Simply renaming the file report.docx to report.pdf. Will it open? NO!

CSS

Cascading Style Sheets (CSS)

- Defines the **style** of HTML
  - "make all text green"
  - "give the first image a red border"
- The "adjectives" of a webpage



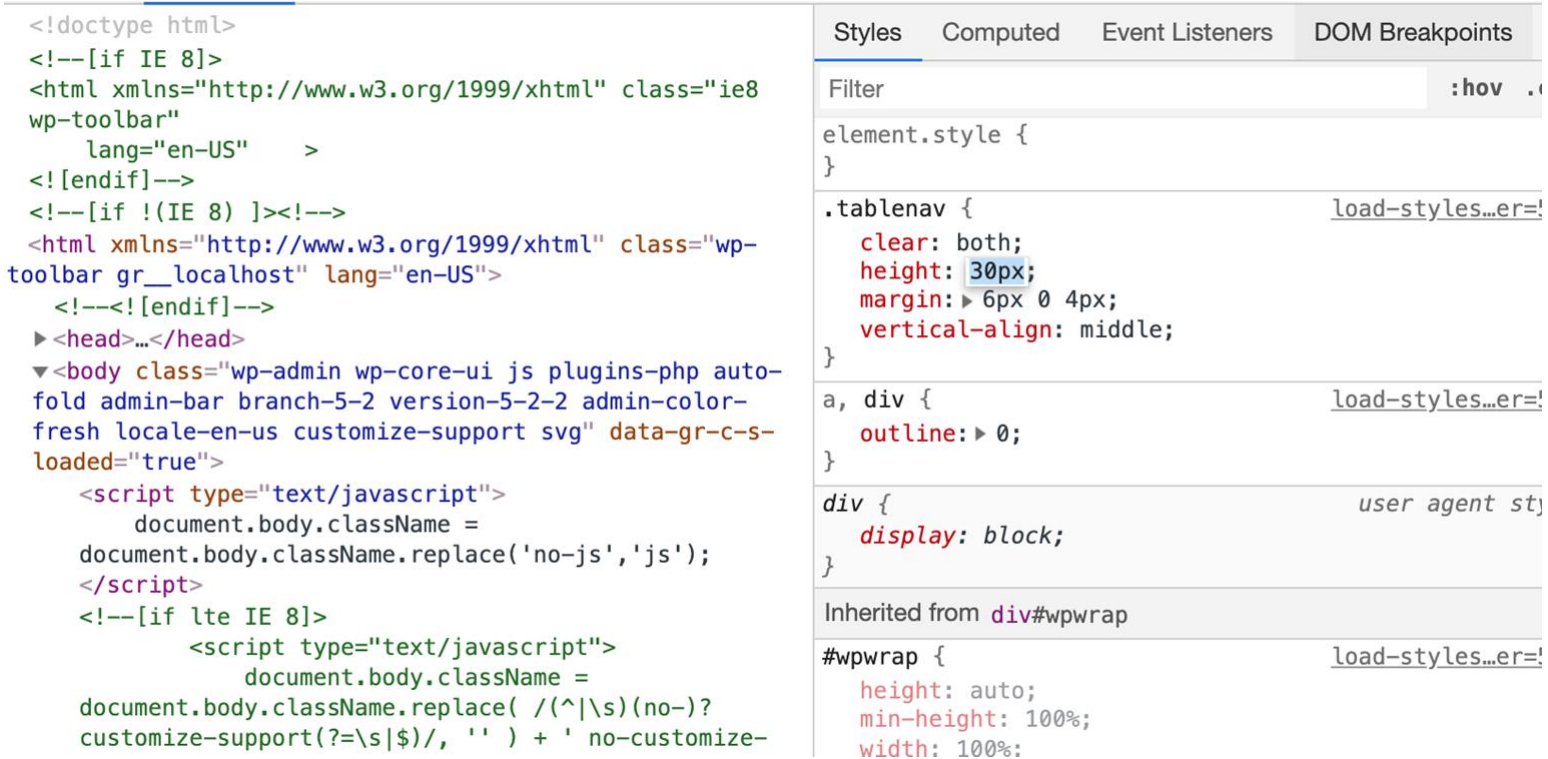
Without CSS



With CSS



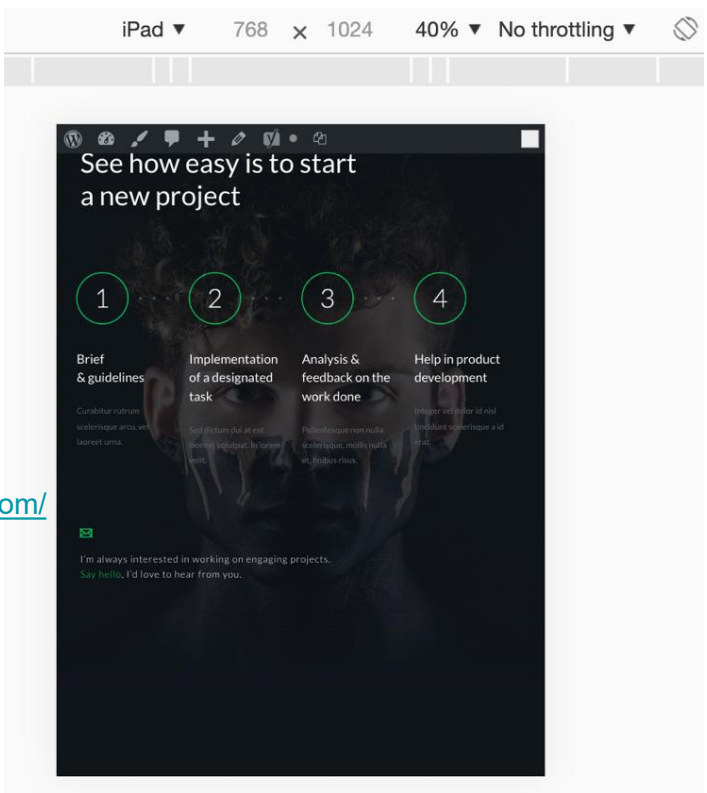
INSPECTOR MODE

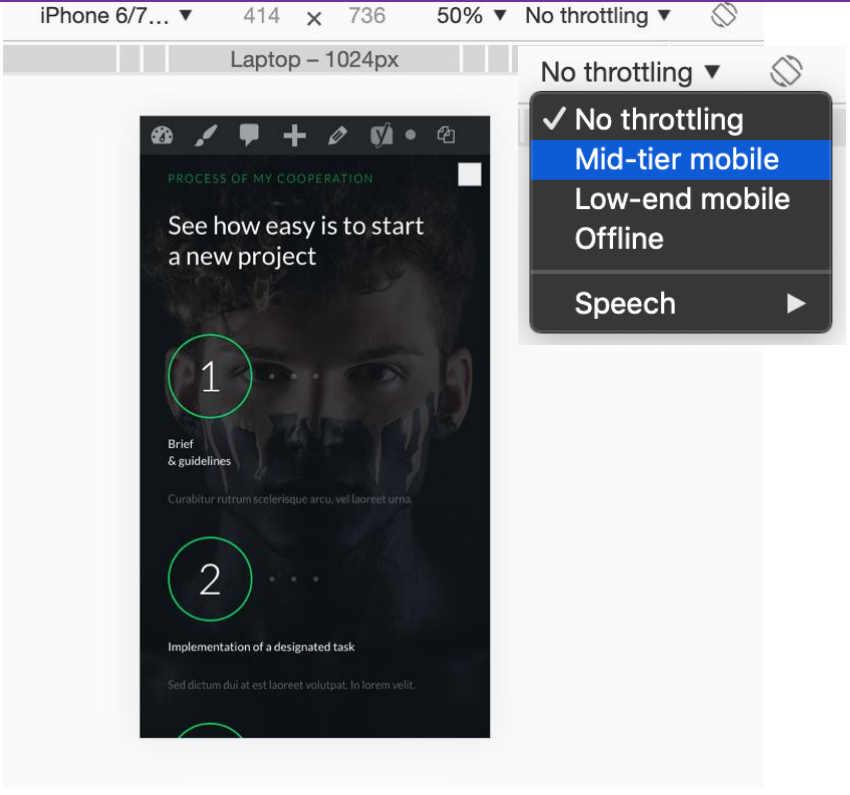


DEVICE MODE

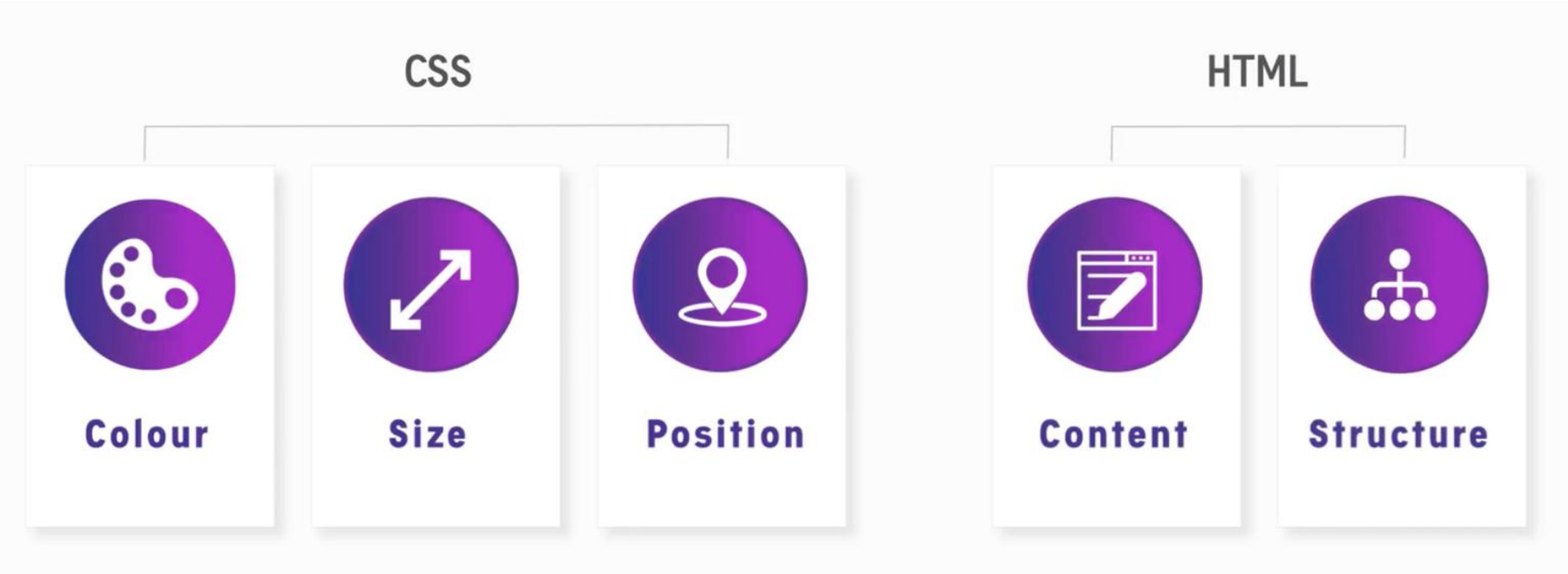
The device mode allows you to toggle devices for preview and testing

Alternatively
https://tools.pingdom.com/





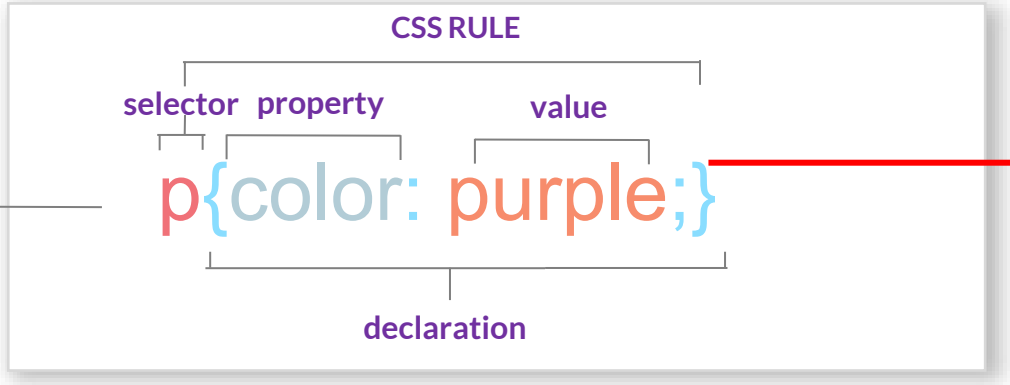
CSS



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CSS Rule

selector { property: value;}



curly braces, all styles inside these will apply to the HTML elements that belong to the selector

CURLY BRACES
!NOT angled brackets
<>

Rule for paragraph selector.
Affects all paragraphs in the webpage

This is a plain old paragraph. However, we can jazz it up with some CSS. Let's get those selectors working and firing away.

Examples : wk02-simple-css

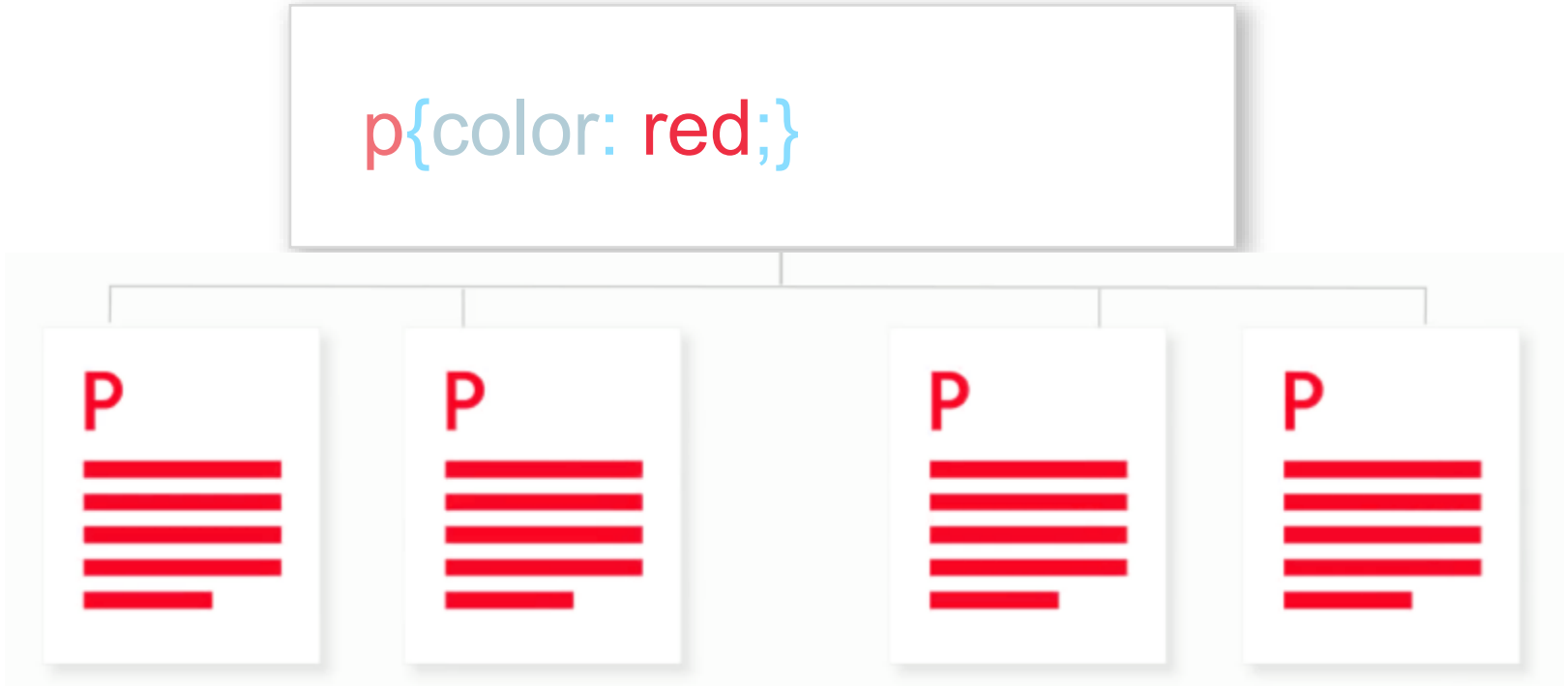
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CSS Rule



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CSS Rule



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CSS: Inline Style (Method 1)

```
<p style="text-align:
left;">Just a sentence</p>
```

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CSS: Style Element (Method 2)

```
<head>
    └   <style>
        *, body{
        font-family: Arial, Helvetica, sans-serif;
        }
        p{color: green;}
        </style>
</head>
```

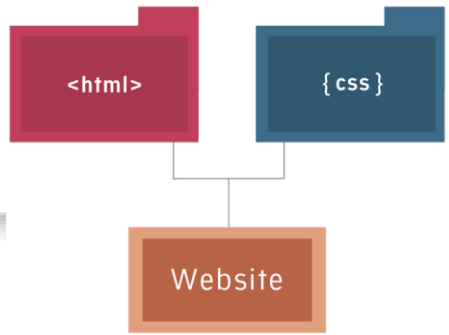
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CSS: Link Element/External Style Sheet (Method 3)

MOST PREFERRED METHOD

```
<head>
    <link rel="stylesheet"
    href="basic.css" type="text/css">
</head>
```

- The href attribute points to the external file.
- The CSS file must have a .css extension.
- rel attribute defines the relationship between the HTML and CSS files.
- rel="stylesheet" means that this file imports the style for the current page.



Ways to include styles in HTML

INLINE STYLE

- Apply directly to individual elements with a HTML element's style attribute
- Targets only that element

<style>

- Should appear as a child of the <head> element on your page
- Styles all occurrences of a target that are on the page

<link>

- Set of style rules defined in an external file
- Should appear as child of the <head> element of your page
- Styles all occurrences of a target that are on a page
- Central external point for styling many pages

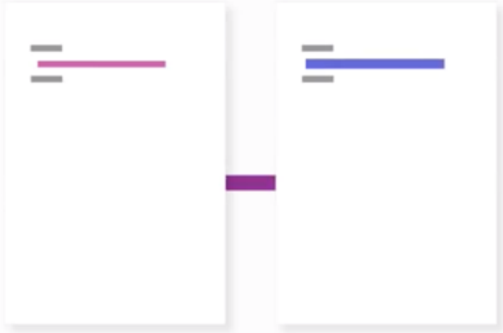
@import

- Set of style rules defined in an external file
- Should appear as a first rule in an external stylesheet
- Used to chain multiple external files together



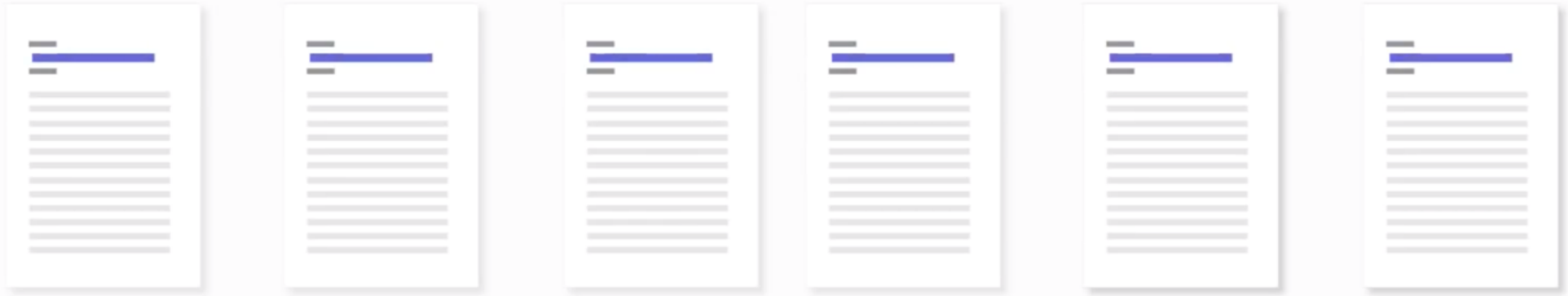
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Link Element



```
<head>
    <link rel="stylesheet" href="basic.css" type="text/css">
</head>
```

Link
Stylesheet



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COLORS

```
h1{
    color: purple;
}

h2{
    color: cornflowerblue;
}

h3{
    color: darkorchid;
}
```

```
<h1>This is h1</h1>
<h2>This is confidence</h2>
<h3>Mini h3 here</h3>
```

This is h1

This is confidence

Mini h3 here

aliceblue	antiquewhite	aqua	aquamarine	azure	beige
bisque	black	blanchedalmond	blue	blueviolet	brown
burlywood	cadetblue	chartreuse	chocolate	coral	cornflowerblue
cornsilk	crimson	cyan	darkblue	darkcyan	darkgoldenrod
darkgray	darkgreen	darkkhaki	darkmagenta	darkolivegreen	darkorange
darkorchid	darkred	darksalmon	darkseagreen	darkslateblue	darkslategray
darkturquoise	darkviolet	deeppink	deepskyblue	dimgray	dodgerblue
firebrick	floralwhite	forestgreen	fuchsia	gainsboro	ghostwhite
gold	goldenrod	gray	green	greenyellow	honeydew
hotpink	indianred	indigo	ivory	khaki	lavender
lavenderblush	lawngreen	lemonchiffon	lightblue	lightcoral	lightcyan
lightgoldenrodyellow	lightgray	lightgreen	lightpink	lightsalmon	lightseagreen
lightskyblue	lightslategray	lightsteelblue	lightyellow	lime	limegreen
linen	magenta	maroon	mediumaquamarine	mediumblue	mediumorchid
mediumpurple	mediumseagreen	mediumslateblue	mediumspringgreen	mediumturquoise	mediumvioletred
midnightblue	mintcream	mistyrose	moccasin	navajowhite	navy
oldlace	olive	olivedrab	orange	orangered	orchid
palegoldenrod	palegreen	paleturquoise	palevioletred	papayawhip	peachpuff
peru	pink	plum	powderblue	purple	rebeccapurple
red	rosybrown	royalblue	saddlebrown	salmon	sandybrown
seagreen	seashell	sienna	silver	skyblue	slateblue
slategray	snow	springgreen	steelblue	tan	teal
thistle	tomato	turquoise	violet	wheat	white
whitesmoke	yellow	yellowgreen			

Base 142 Colors

http://www.colors.commutercreative.com/

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COLORS

Hexadecimal mode

#+ String of 6 hexadecimal numbers (from 0-F) 16.7 million colors

#ffffff – white | #000000 - black

```
h1{
    color: #000000;
}

h2{
    color: #4b0082;
}

h3{
    color: #ff1493;
}
```

```
<h1>This is h1</h1>
<h2>This is confidence</h2>
<h3>Mini h3 here</h3>
```



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COLORS

RGB

3 Channels: Red, Green and Blue (Each ranges from 0 –

```
h1{
    color: rgb(0,0,0);
}

h2{
    color: rgb(75,0,130);
}

h3{
    color: rgb(255,20,147);
}
```

```
<h1>This is h1</h1>
<h2>This is confidence</h2>
<h3>Mini h3 here</h3>
```

This is h1

This is confidence

Mini h3 here

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COLORS

RGB**A**

Just like RBG, but with an alpha(transparency) channel. Ranges from 0.0-1.0

```css
h1{
    color: rgba(11, 99, 150, 1);
}

h2{
    color: rgba(11, 99, 150, 0.6);
}

h3{
    color: rgba(11, 99, 150, 0.2);
}
```

```html
<h1>This is h1</h1>
<h2>This is confidence</h2>
<h3>Mini h3 here</h3>
```

<h1>This is h1</h1>

<h2>This is confidence</h2>

<h3>Mini h3 here</h3>

Colors Again

The "color" property **changes the text color.** Colors can be specified either by name, for the most common colors, or by hexadecimal value.

color: red;
color: #ff0000;
color: rgb(255, 0, 0);

```
<p style="color: red">red text</p>
```

red text

The "background-color" property changes the background color. Besides the BODY element, all elements default to a transparent background.

background-color: black;
background-color: #000000;
background-color: rgb(0,0,0);

```
<p style="background-color: red; color: white;">red text</p>
```

red
text

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30 minutes CA 1

CHALLENGE: Fun with Styles

DIRECTIONS - Use Visual Studio Code

1. Create your 1st website
2. Create the external CSS file “style.css”
3. Add the css file using the <link> method
4. Add a few CSS properties (color, background-color, font-family, text-align) to the content created in the last activity
5. Open up the HTML file in Google Chrome

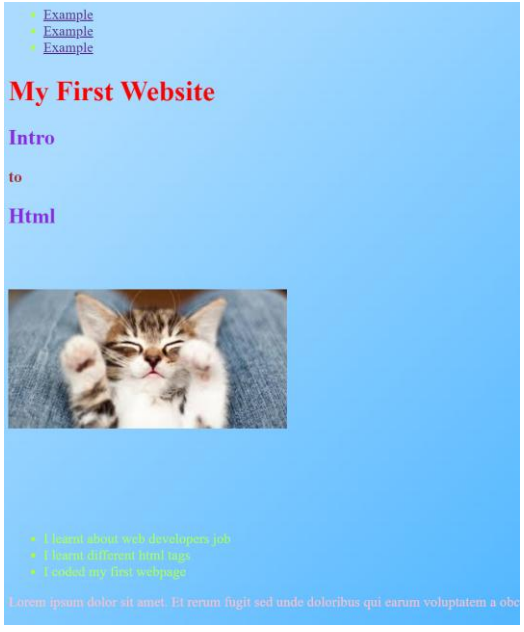
Submit a html titled: “Wk02-studentid-first-website.html”
Submit a CSS file titled: “style.css”

E.g Ik02-16012010D-UncleRoger-first-website.html
*Submit the file in **Brightspace***

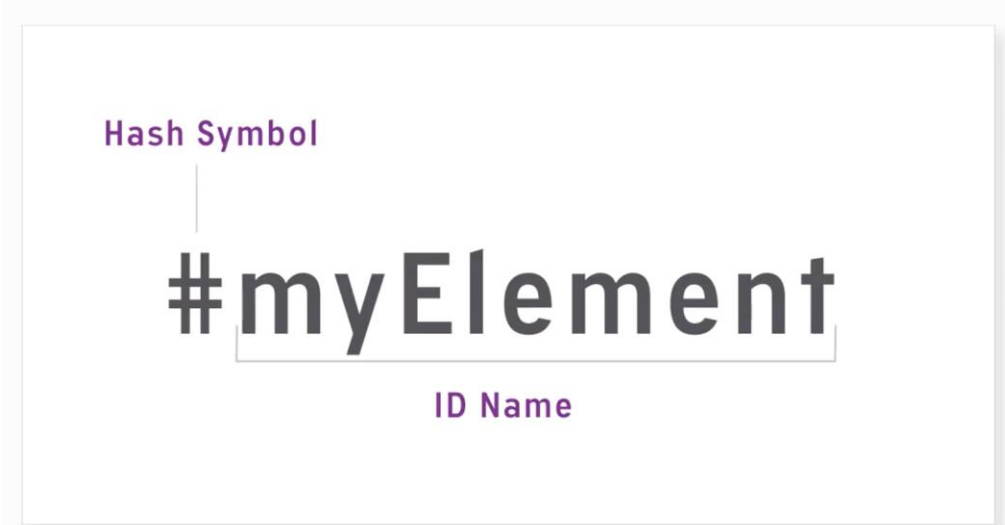
https://code.visualstudio.com/

Some KickAss Visual Studio Code Extensions
Live-Server, Beautify, HTML Boilerplate, HTML Snippets

Create a html with your own contents. You must have lists, h1,h2 and an img minimally. You can include paragraphs and others if you wish to. Final sample output shown below.



ID Attribute



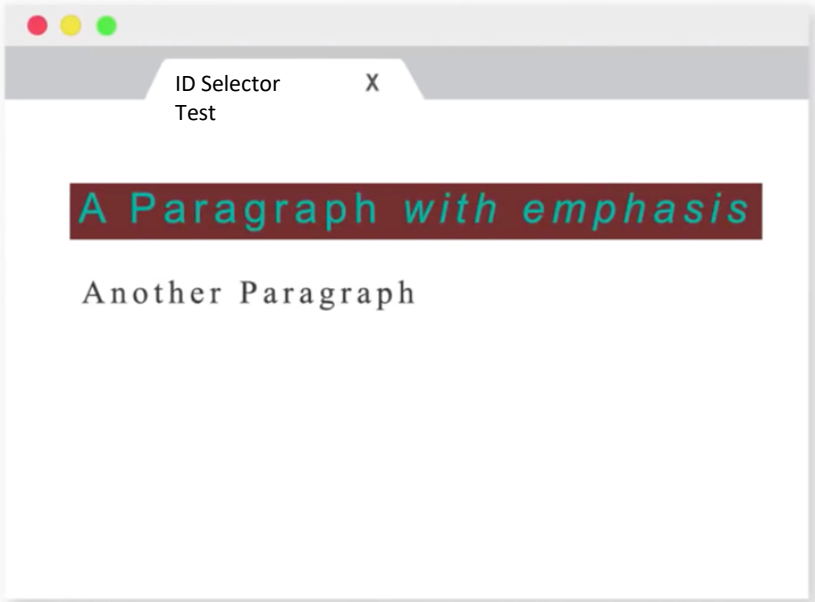
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ID Attribute

```
<!DOCTYPE html>
<html>
<head>
  <title>ID Selector Test</title>
  <style>
    #myElement{font-family:Arial, Helvetica, sans-serif;color:
#23FBFF;background-color: #6E3B04;}
  </style>
  </head>
<body>
  <p id="myElement">
    A paragraph <em> with emphasis</em>
  </p>

  <p>
    Another Paragraph
  </p>
</body>
</html>
```





Class Attribute



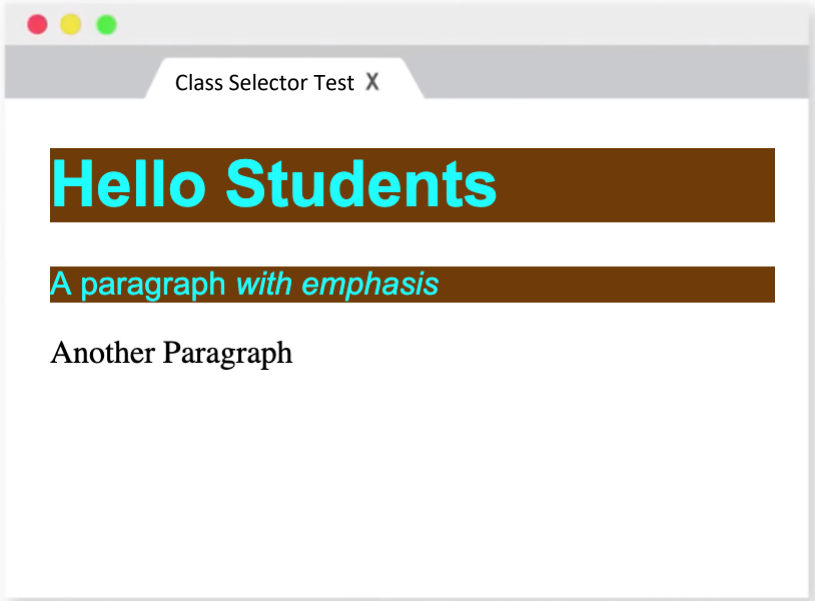
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Class Attribute

```
<!DOCTYPE html>
<html>
<head>
  <title>Class Selector Test</title>
  <style>
  .myClass{font-family:Arial, Helvetica, sans-serif;
  color: #23FBFF;background-color: #6E3B04;}
  </style>
</head>
<body>
  <h1 class="myClass">Hello Students</h1>
  <p class="myClass">
  A paragraph <em> with emphasis</em>
  </p>

  <p>Another Paragraph</p>
</body>
</html>
```





Class Attribute

```
In CSS, an ID is #

// change the color of the element with id "main"
#main {
   color: black
}
// change the color of the element with class special
.special {
   color: black
}
You can specify a HTML tag and a class to be more specific
// for all <p> with the class special
p.special {

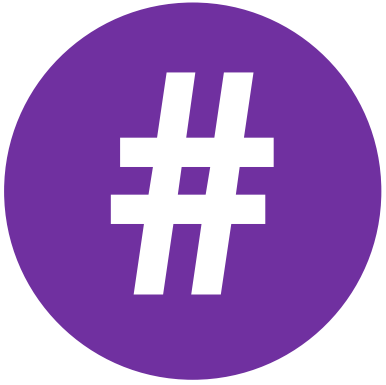
}
You can target a children element that is in a parent
element
// all <a> inside the ID "main" html element
#main a {

}
```



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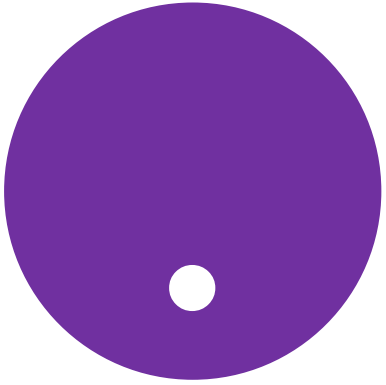
ID Attribute - TARGETING AN INDIVIDUAL



- #header { } Selects any element with the **id** “header", e.g.
- Unique to that element
- Hash symbol
- The "**#**" is how you tell CSS "this is an id.”
  - Selector name matches element ID value, prefixed with # symbol

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Class Attribute - TARGETING A GROUP



- .warning { color: red; } Selects any element with the **class name** “warning”, e.g.
- Used to style an element grouping
- Targets one or more element's class attribute
- Elements can be of different type
- The "." is how you tell CSS "this is a class name."
- An element can have only 1 id but multiple classes, so you can take advantage of that for greater flexibility.
  - Eg hi -> hi

  aka class chaining

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Pseudo Classes

A set of "pseudo classes" can style anchor elements depending on their state.

```
a:link { /* unvisited link */ color: red; }
a:visited { /* visited link */ color: blue; }
a:hover { /* moused over link */ color: green; }
a:active { /* current link */ color: purple; }
a:focus { /* focused link */ color: purple; }
```

Some rules to follow

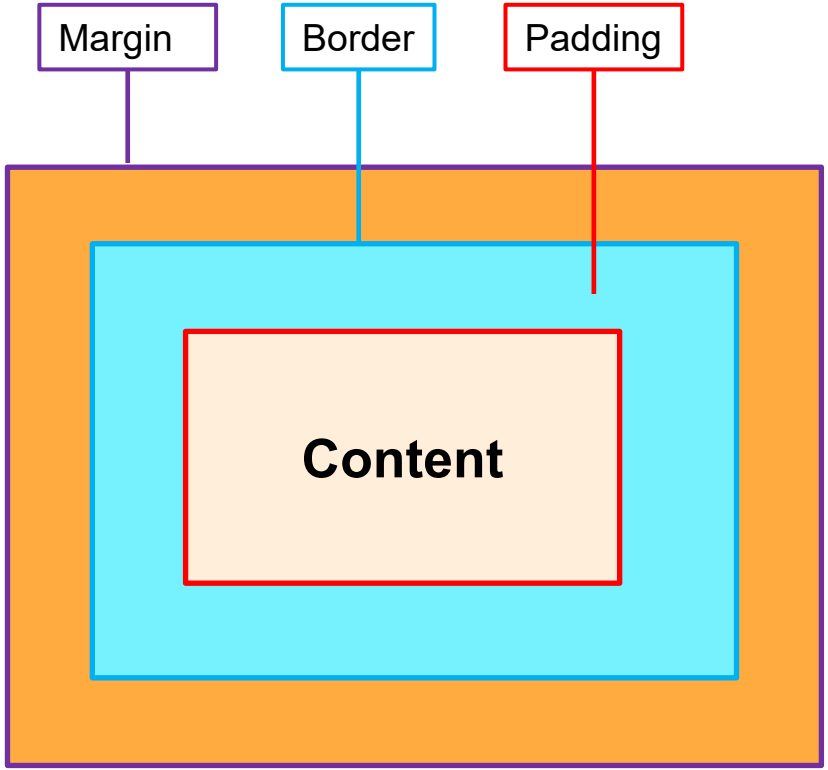
Some rules to follow when making IDs and class names:

- **Describe the content**, not the presentation ("warning", not "redbox").
- Use **all lowercase, and hyphens when needed for readability** ("header-info", not "headerInfo").
- Use hyphens to show that a class or ID is part of something else. (e.g. "footer", "footer-copyright", and "footer-logo").
- Preferably **NO** camel case



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Box Model



What is the CSS Box Model?

The CSS box model is a fundamental concept for web layout and design. It describes how elements are structured and spaced on a webpage.

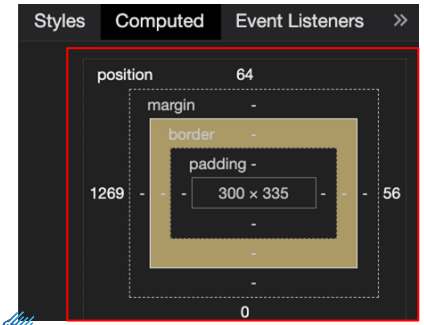
Components of the Box Model

- **Content:** The actual text, image, or media inside the element.
- **Padding:** Space between the content and the border (inner space)
- **Border:** The line that wraps around the padding and content.
- **Margin:** Space outside the border, separating the element from others (outer space)

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Box Model





Examples : wk02-simple-boxmodel

Why is it Important?

Understanding the box model is essential for creating precise layouts, controlling spacing, and debugging visual issues in web pages.

Example CSS:

```
div {
  margin: 10px;        /* Space outside the box */
  padding: 10px;       /* Space inside the box */
  border: 2px solid #000; /* Border around the box */
}
```

Tips:

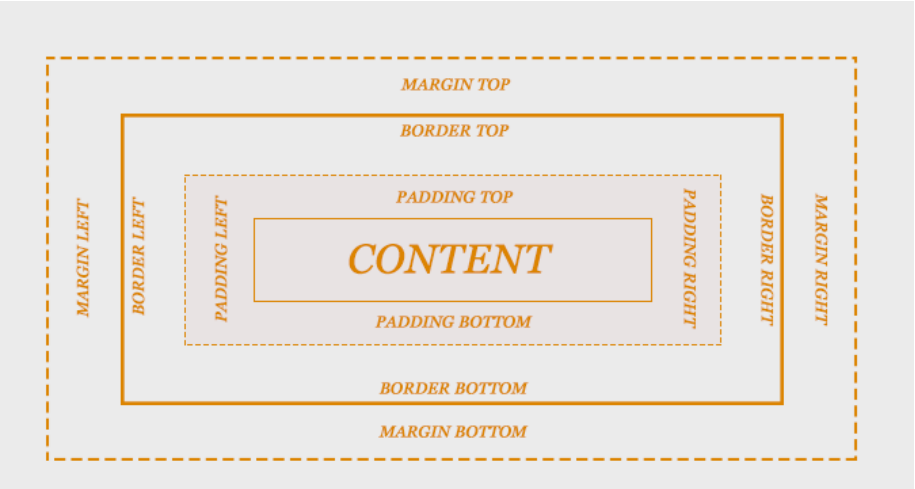
Margin and padding can be set for each side: top, right, bottom, left.

Example of Shorthand:

```
margin: 10px 10px 5px 5px; /* top right bottom left */
```

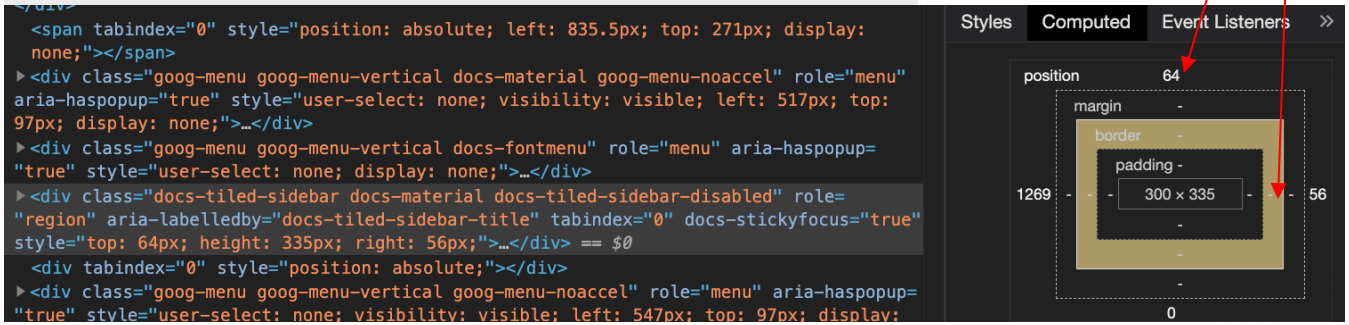
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Box Model



Try it out!

Now try right click and inspect your html code.
Navigate to any section of content and view it's computed mode.
Click on any area of the box and modify the settings. See and observer what happens.



Examples : wk02-simple-boxmodel

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20 minutes CA 2

Applying Class & ID

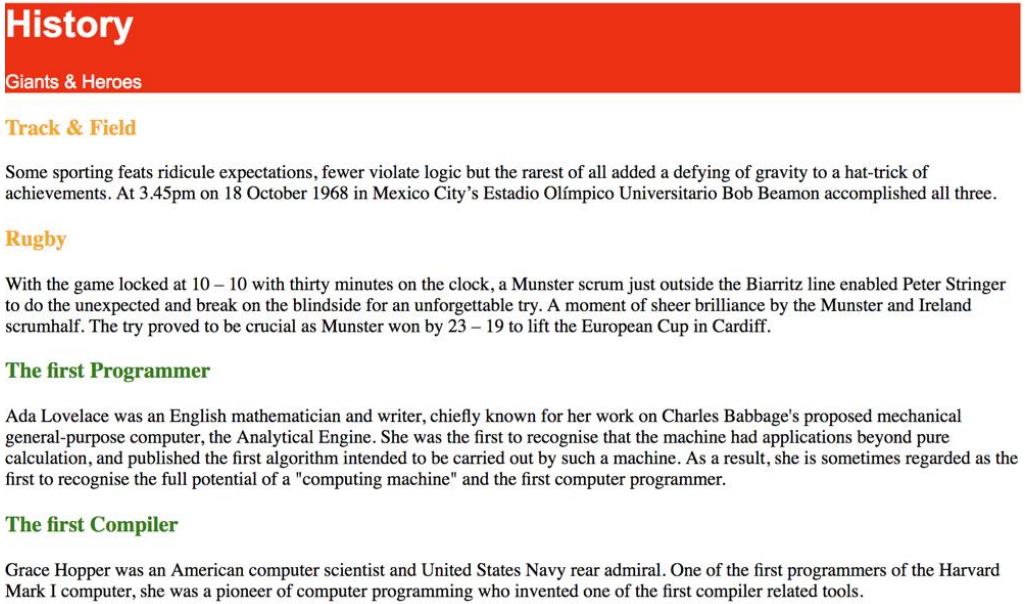
1. Work on the files in wk02-css-id-class
2. Apply the #heading ID to ***one*** appropriate element to affect the result to look like the one on the right
3. Apply the class styles to the appropriate elements to affect the result

Submit a html titled: “Wk02-studentid-studentname-css-id.html”

E.g Wk02-16012010D-UncleRoger-css-id.html
Submit the file in ***Brightspace***

https://code.visualstudio.com/

Some KickAss Visual Studio Code Extensions
Live-Server, Prettier, HTML Boilerplate, HTML Snippets



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Fonts

in CSS

A look at the possible font properties

```
.theme-font{
    font: 1.2em "Fira Sans", sans-serif; /* all properties*/
    font-family: "Helvetica", Arial, sans-serif;
    font-size: 19px; /* px,em,%, xx-small to xx-large */
    font-style: italic; /* normal, italic, oblique */
    font-variant: normal /* normal, small-caps */;
    font-weight: bold; /* bold, bolder, lighter, normal, 100-900 */
    line-height: 1.5em; /* line height b/w two lines of text */
    text-align: center; /* alignment of text within the box */
}
```

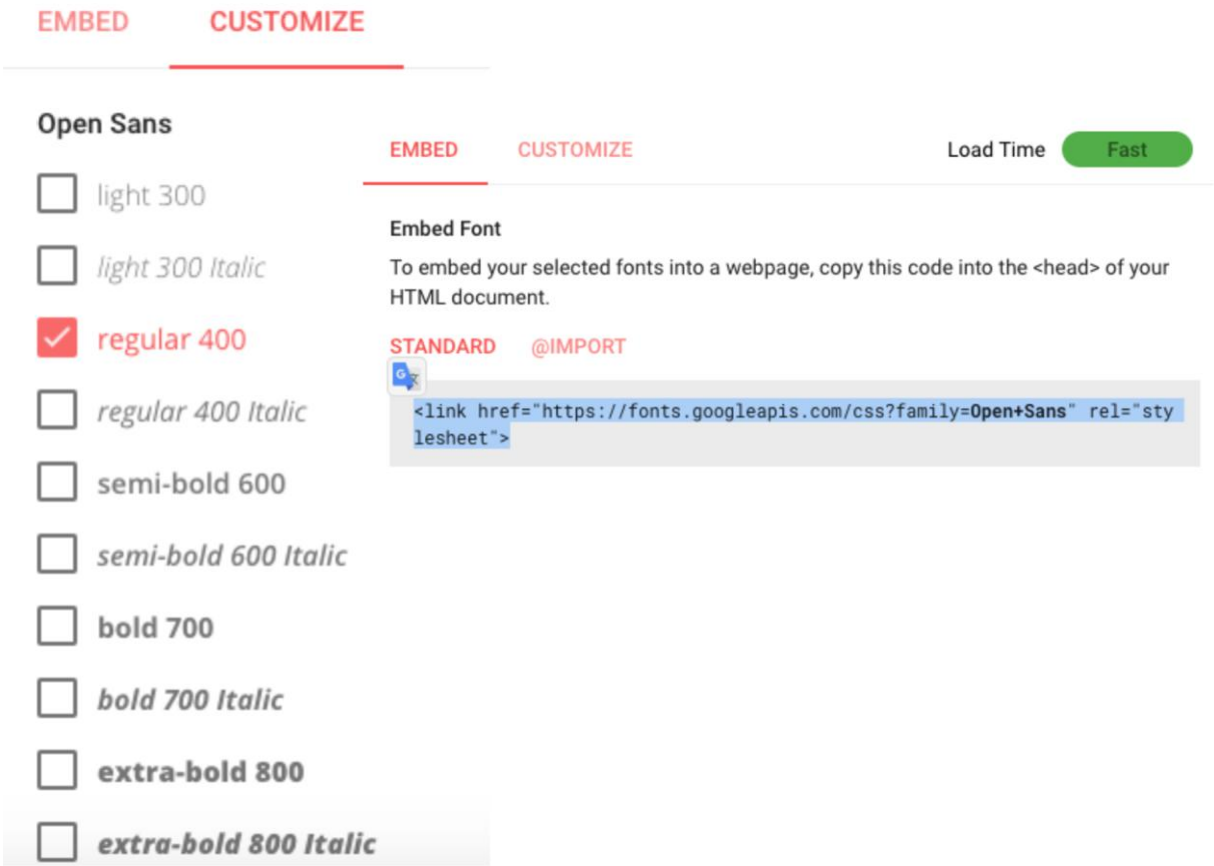
Fonts with a space in their names are to use **quotes**

"Fira Sans"

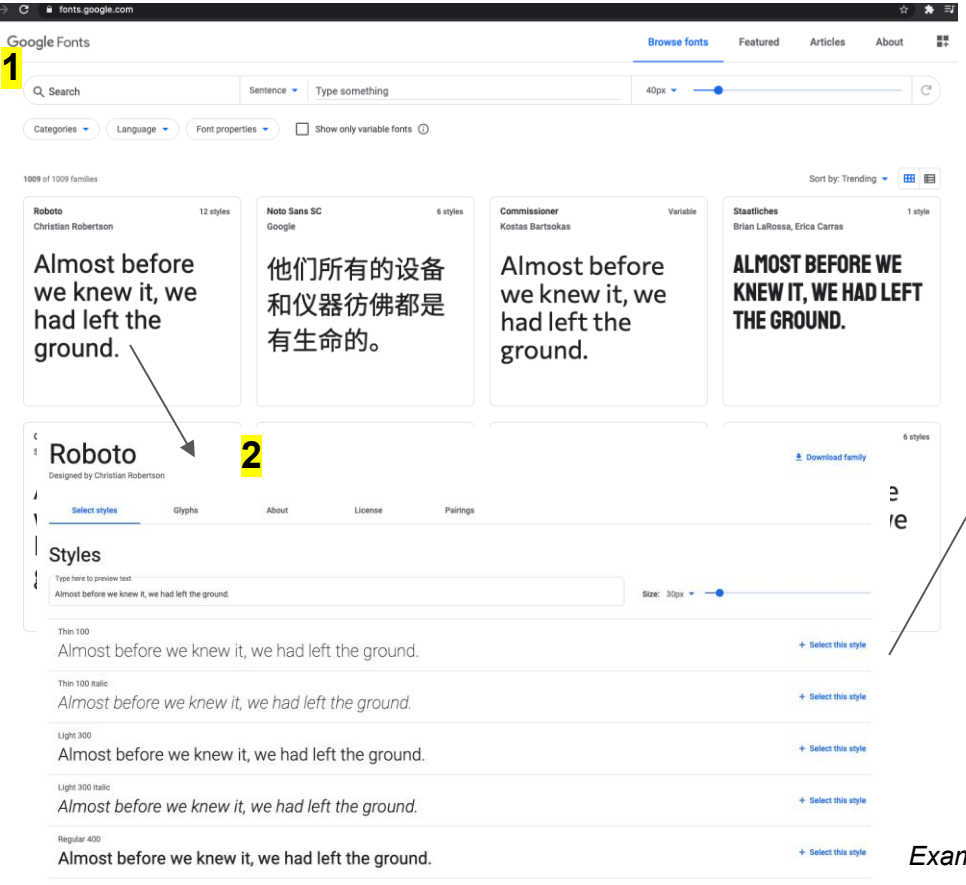


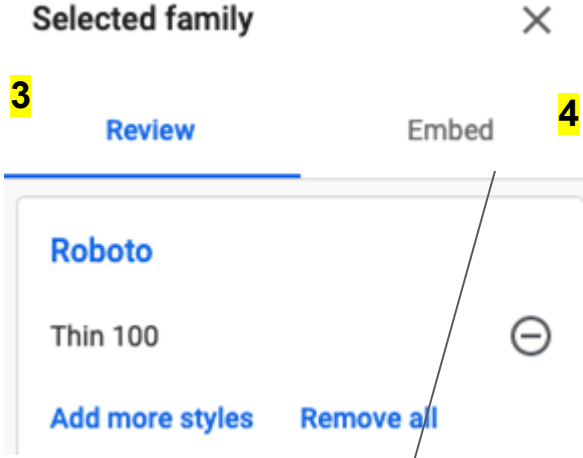
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Google Fonts



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5

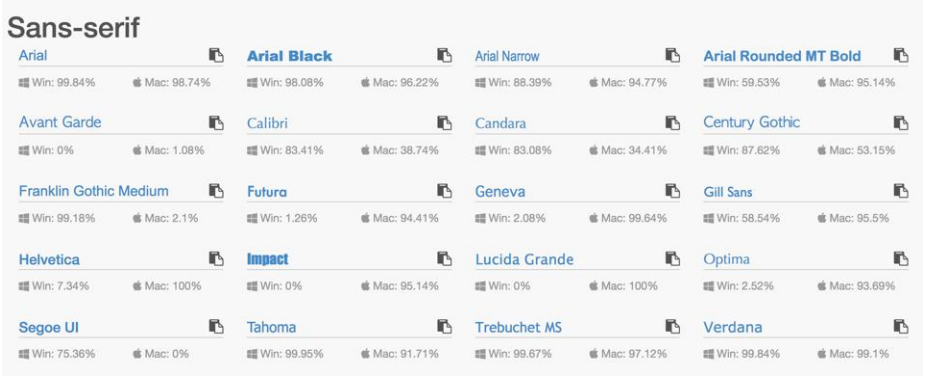
Place link into the <head> element
Add on CSS rule to your CSS styles
Now test it out!

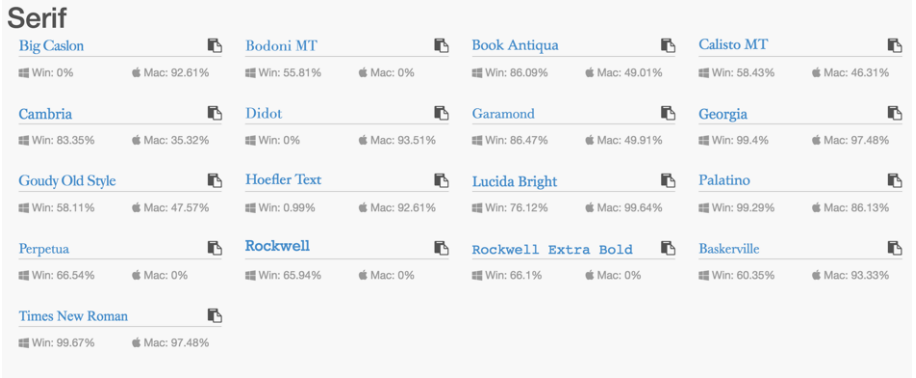


Examples : wk02-simple-googlefonts

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FONT SUPPORT LEVELS





Source: https://www.cssfontstack.com/

```
    <style>
body {background-color: lightyellow;}
h1 {color: black;font-family: Calibri}
p {font-family: Calibri;font-size: 15px;}
h2 {color: black;font-family: Calibri}

li {font-family: Calibri;}
table{font-family: Calibri;}
</style>
```

Student's code from Recipe. Note there's no backup font

Arial CSS Font Stack

Stack Name

Arial

Compatability

Win: 99.84% Mac: 98.74%

Fonts in Stack:

Arial, Helvetica Neue, Helvetica, sans-serif

Font: Arial Full Stack | Size: Select One: | Weight: Select One: | Style: Select One:

Variant: Select One: | Line Height: Select One:

Apply To

Header | Save Heading Setting

HTML:

CSS:

```
h1 {
        font-family: Arial, "Helvetica Neue", Helvetica, sans-serif;
        font-size: 24px;
        font-style: normal;
        font-variant: normal;
        font-weight: 500;
        line-height: 26.4px;
}
```

Preview Your Fonts

Your headline is in Arial

This is a sub heading in Arial.

This paragraph is in Arial. Keep reading for how to use the buttons to the left. Change your settings, and choose what section you want the font to apply to. Once you are pleased with the settings for that section you can click save. Your selections will remain as you look for another font for another section of this preview.

Another sub heading in Arial.

The setting for a paragraph continues down here. There is a blockquote next to it. You may want to make that stand out. The setting for a paragraph continues down here. There is a blockquote next to it. You may want to make that stand out. The setting for a paragraph continues down here. There is a blockquote next to it. You may want to make that stand out.

> Make other Arial text stand out!

```
/* This text is in Arial */
.class {
      font-family: Arial,Helvetica Neue,Helvetica,sans-serif;
}
```

Golden Ratio | Reset

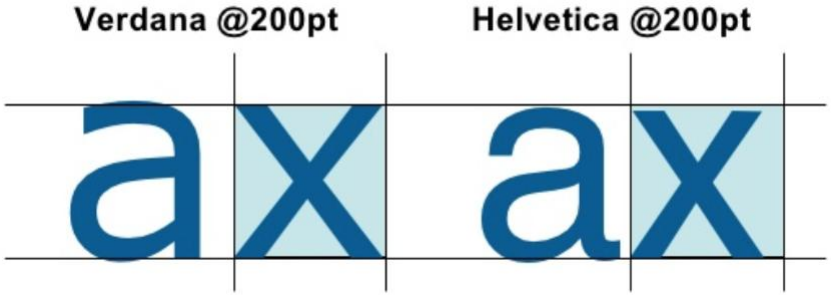
Apply Golden Ratio to font sizes and line height based on paragraph font size.

Source: https://www.cssfontstack.com/

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FONT STACKING

- Think about fallback fonts
- Platform usage (Windows, Mac, Linux etc)
- Font aspect ratio
  - If you use fonts with different aspect ratios, Some people may see your site with smaller fonts



Example of difference in Font aspect ratio

```
body{
   font-family: Verdana, Arial, sans-serif
}
```

Larger aspect ratio

Smaller aspect ratio

Fonts

The "font-size" property specifies the size of a font. It can be specified as a fixed size in various units, a percentage, or as a predefined keyword.

```
font-size: 1.5em;
font-size: 12px;
font-size: 100%;
font-size: larger;
```

The "em" unit lets you set the size of the text relative to the text around it. This makes the page resize nicely in proportion if the user changes their default font-size. The default size is "1em" or 16px.

```
p { font-size: 0.9em; }
strong { font-size: 1.5em; }
```



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Fonts

The "px" unit lets you size font in terms of pixels, which is the unit also used to size images and other elements. It is easier to understand than em, but doesn't work as well when printing or resizing.

```
h2 { font-size: 17px; }
```

There are various keywords that can be used if you're not as worried about the precise sizing: xx-small, x-small, small, medium, large, x-large, xx-large.

```
p.footnote { font-size: small; }
```

The size can also be specified as a percentage, which works similar to "ems", and can be used in conjunction with other units.

```
body { font-size: 12px; }
h1 { font-size: 200%; }
h1 a { font-size: 75%; }
```



Fonts

The "font-style" property specifies the font style of the text, either "normal" by default or "italic".

```
font-style: italic;
```

The "font-weight" property specifies the thickness of the font.
The default is "normal" and the typical override is "bold". You can also specify "bolder", "lighter", or a number from 100 to 900.

```
font-weight: bold;
```

Additional Reading https://developer.mozilla.org/en-US/docs/Web/CSS/font

A "shorthand" property in CSS lets you specify multiple properties in one property, for conciseness purposes. Instead of specifying each "font-" property separately, you can **bundle them up** in one "font" property.

```
table { font-weight: bold;
        font-style: italic;
        font-size: 10px;
        font-family: sans-serif; }
```

Those four rules can be written as:

```
table { font: italic bold 10px sans-
serif; }
```



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20 minutes CA 2

Importing Google Fonts

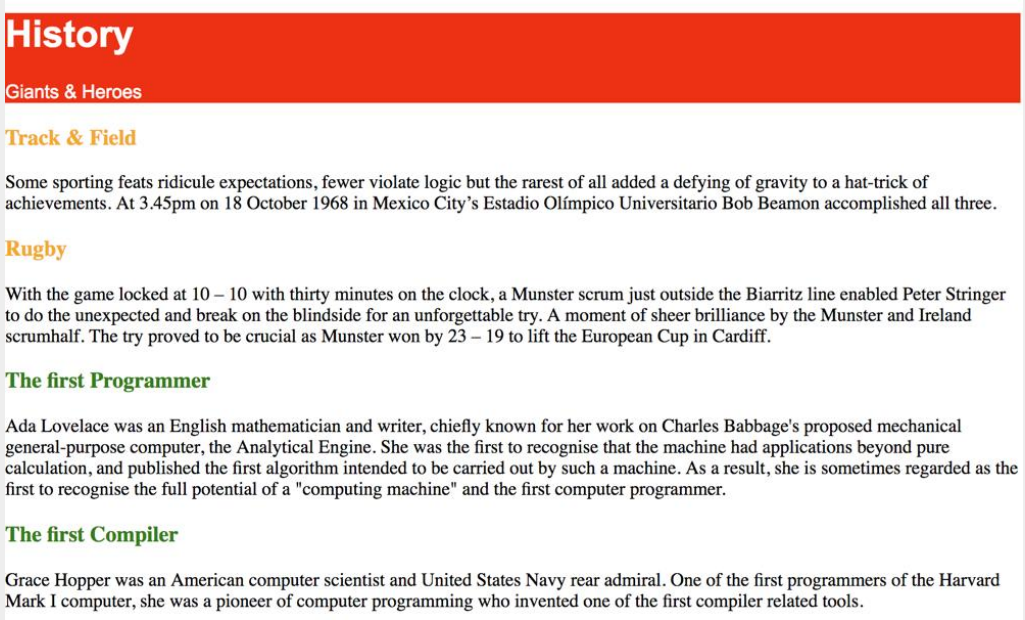
1. Duplicate your previous CA and continue
2. Go to fonts.google.com
3. Select the fonts you want to implement
4. Choose at least 2 Google fonts that you can apply for headings and the paragraphs. You can use the same stylesheet

CA2 Submit a html titled: “Wk02-studentid-studentname-googlefonts.html”

E.g Wk02-16012010D-UncleRoger-googlefonts.html
*Submit the file in **Brightspace***

https://code.visualstudio.com/

Some KickAss Visual Studio Code Extensions
Live-Server, Prettier, HTML Boilerplate, HTML Snippets



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Submit all 3 CAs

ZIP UP ALL your html and submit
E.g Wk02-16012010D-UncleRoger.zip

*Submit the zipped file in **Brightspace***

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WEEK 3

(FED) 1.2

More CSS
CSS POSITION
Media Queries
Flexbox

2025

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FRONT END DEVELOPMENT
IS A ~~PAIN~~ DO-ABLE
I SURVIVED…



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BYE

Watch these videos!





Don’t just read w3schools
Read this :)
https://developer.mozilla.org/en-US/docs/Web/CSS